

Analysis and Prediction of Diamond Prices

Using Multiple Linear Regression

Under the supervision of Dr. Tuhinsubhra Bhattacharya

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PROJECT REPORT

In partial fulfillment of the requirements
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(Maulana Azad College)



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CERTIFICATE

This is to certify that the project paper titled “*Analysis and Prediction of Diamond Prices*” has been submitted by **Supriyo Saha** in partial fulfillment of the requirements for the B.Sc. (Honours) degree in Statistics from University of Calcutta (Maulana Azad College).

The project work presented in this report has been conducted under my guidance and supervision. I can confirm that the results and findings presented in this report are the outcome of the student’s genuine efforts and the analysis conducted during the specified period.

Furthermore, to the best of my knowledge, the results and findings reported in this project have not been submitted for the award of any other degree or diploma in any other academic institution.

I recommend this project paper for evaluation and fulfillment of the academic requirements.

Date: _____

Place: Kolkata

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DECLARATION

I, **Supriyo Saha**, student of B.Sc. (Honours) in Statistics at Maulana Azad College, affiliated with the University of Calcutta, hereby declare that the project report entitled “*Analysis and Prediction of Diamond Prices*” submitted for the degree of B.Sc. (Honours) in Statistics is an authentic record of original work carried out by me under the supervision of **Dr. Tuhinshubra Bhattacharya**.

I further declare that this work is original and has not been submitted previously, in part or in full, to any other University or Institution for the award of any degree, diploma, or fellowship.

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1 Introduction

Machine Learning (ML) is a branch of artificial intelligence that allows computers to learn patterns from data and use those patterns to make predictions or decisions without being explicitly programmed for every individual task.

Machine learning methods can broadly be divided into three major categories:

1.1 Supervised Learning

In supervised learning, the model is trained using a dataset in which the target or response variable is known.

The main types of supervised learning are:

- **Regression:** Used when the response variable is numerical and continuous.
- **Classification:** Used when the response variable belongs to different categories.

Examples of supervised learning algorithms include:

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- Support Vector Machines

1.2 Unsupervised Learning

In unsupervised learning, the target variable is not provided. The algorithm attempts to discover hidden patterns or structures within the data.

Common examples include:

- K-means clustering
- Hierarchical clustering
- Principal Component Analysis (PCA)

1.3 Reinforcement Learning

Reinforcement learning is a type of machine learning in which an agent learns by interacting with an environment. The agent receives rewards or penalties depending on its actions and attempts to learn a strategy that maximizes the total reward.

1.4 Machine Learning Method Used in This Project

The present study uses **supervised learning**, specifically **multiple linear regression**.

The response variable is:

Diamond Price

and the explanatory variables considered in the analysis are the physical measurements and quality characteristics of the diamonds, including carat, depth percentage, table percentage, length, width, depth, cut, colour, and clarity.

The regression model was fitted in R using:

```
1 model <- lm(price ~ ., data = x)
```

This corresponds to the multiple linear regression analysis performed on the supplied diamond dataset.

2 Description of the Data

2.1 Dataset

The dataset used in this project contains information about diamonds and their physical and quality-related characteristics. The analysis focuses particularly on understanding the factors associated with diamond price.

The dataset used for the analysis contains:

- **Number of observations:** 53,917
- **Number of variables:** 10
- **Response variable:** price
- **Software used:** R / RStudio

Dataset Source: <https://www.kaggle.com/datasets/shivam2503/diamonds>

The variables used in the analysis are described below.

Table 1: Description of Variables

Variable	Type	Description
carat	Numerical	Weight of the diamond in carats
cut	Categorical	Quality of the diamond cut
color	Categorical	Colour grade of the diamond
clarity	Categorical	Clarity grade of the diamond
depth_percent	Numerical	Total depth percentage of the diamond
table_percent	Numerical	Width of the top facet relative to diameter
price	Numerical	Price of the diamond
length_mm	Numerical	Length of the diamond in millimetres
width_mm	Numerical	Width of the diamond in millimetres
depth_mm	Numerical	Depth of the diamond in millimetres

The variable names used in the supplied R analysis are:

```

1 carat
2 cut
3 color
4 clarity
5 depth_percent
6 table_percent
7 price
8 length_mm
9 width_mm
10 depth_mm

```

2.2 Summary Statistics

Summary statistics were examined using the `summary()` function in R. The numerical variables considered in the analysis were carat, depth percentage, table percentage, price, length, width, and depth.

The boxplots show that:

- Carat has a positively skewed distribution with several high-valued observations.
- Depth percentage is concentrated around its central values, although some unusual observations are present.
- Table percentage is also concentrated within a relatively narrow range with some outliers.
- Price is strongly positively skewed and contains a number of high-priced observations.

- Length, width and depth show right-skewed distributions and several observations outside the main concentration

Table 2: Summary of Numerical Variables

Variable	Min	Q1	Median	Mean	Q3	Max
Carat	0.20	0.40	0.70	0.7979	1.04	5.01
Depth (%)	43.00	61.00	61.80	61.75	62.50	79.00
Table (%)	43.00	56.00	57.00	57.46	59.00	95.00
Price	326	950	2401	3932.80	5324.25	18823
Length (mm)	3.73	4.71	5.70	5.731	6.54	10.74
Width (mm)	3.68	4.72	5.71	5.735	6.54	10.54
Depth (mm)	1.07	2.91	3.53	3.539	4.04	6.98

2.3 Categorical Variables

Frequency distributions were obtained for the categorical variables *cut*, *color*, and *clarity*.

The barplots show that:

- The *Ideal* cut category has the largest frequency, while *Fair* has the smallest frequency.
- The colour categories are not equally distributed, with *G* having a high frequency.
- The clarity categories show substantial differences in frequency, with *SI1* being one of the most frequent categories.

Table 3: Frequency Distribution of Categorical Variables

Variable	Category	Frequency
Cut	Fair	1609
Cut	Good	4902
Cut	Ideal	21547
Cut	Premium	13779
Cut	Very Good	12080
Color	D	6774
Color	E	9795
Color	F	9538
Color	G	11284
Color	H	8297
Color	I	5422
Color	J	2808
Clarity	I1	738
Clarity	IF	1790
Clarity	SI1	13063
Clarity	SI2	9184
Clarity	VS1	8168
Clarity	VS2	12254
Clarity	VVS1	3654
Clarity	VVS2	5066

The R output gives these frequency counts directly.

3 Objective of the Study

The main objectives of this study are:

1. To understand the structure and characteristics of the diamond dataset.
2. To perform exploratory data analysis using suitable statistical graphs.
3. To study the distributions of the numerical variables using boxplots and histograms.
4. To examine the frequency distributions of cut, colour, and clarity.
5. To investigate the relationships between diamond price and important numerical variables such as carat, length, width, and depth.
6. To measure the correlations among the numerical variables.
7. To construct a multiple linear regression model for predicting diamond price.

8. To evaluate the fitted regression model using appropriate diagnostic plots and statistical measures.

4 Exploratory Data Analysis

4.1 Boxplot Analysis

Boxplots were created for the numerical variables in the dataset: carat, depth percentage, table percentage, price, length, width, and depth.

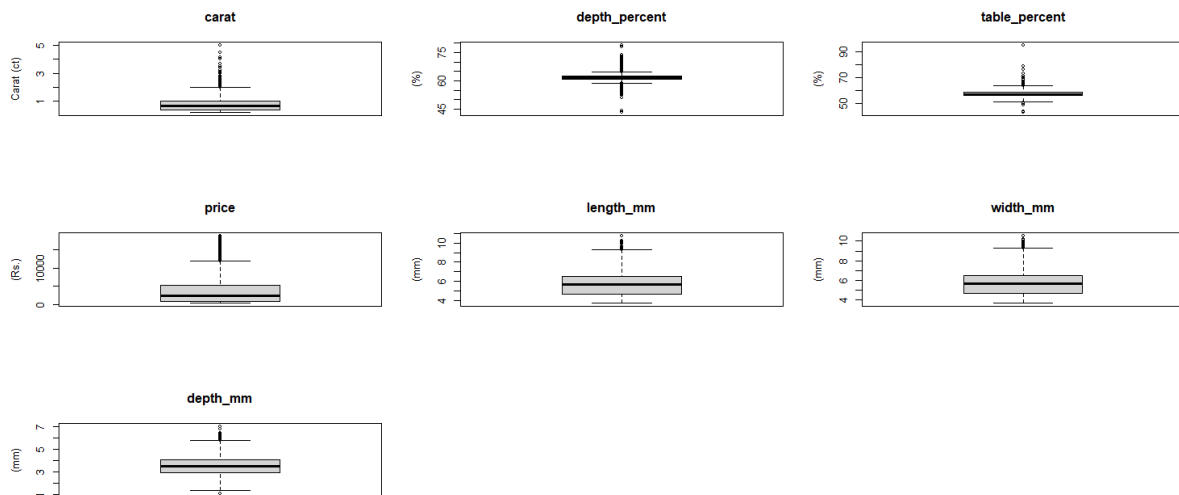


Figure 1: Boxplots of numerical variables

The boxplots provide information about the median, interquartile range, spread, and possible outlying observations.

The *carat* and *price* variables show clear right-skewness and a large number of high-value observations. The dimensional variables also contain several observations outside their main interquartile ranges.

4.2 Barplot Analysis

Barplots were created for the categorical variables *cut*, *color*, and *clarity*.

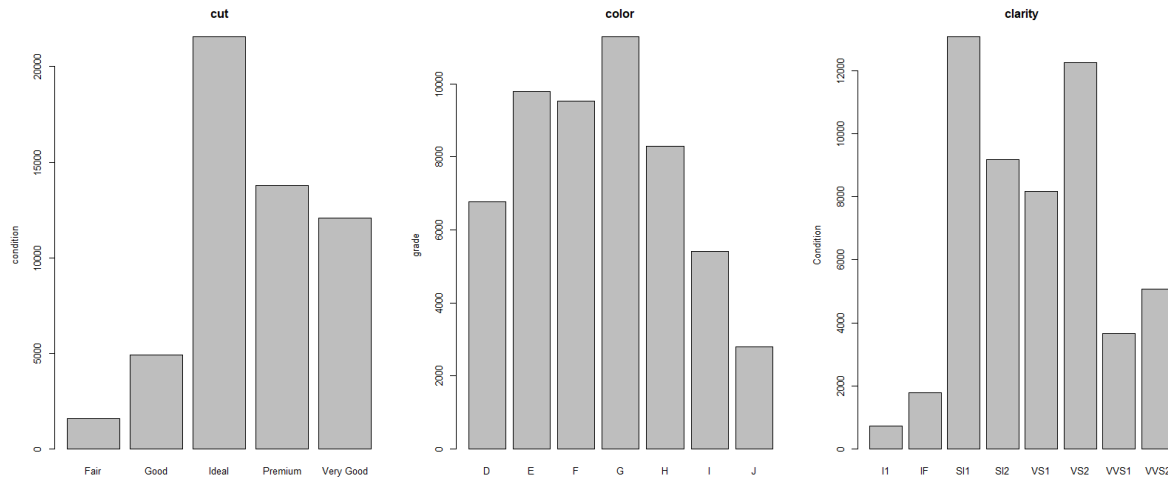


Figure 2: Frequency distributions of cut, colour and clarity

The *cut* barplot shows that Ideal-cut diamonds form the largest group. The colour distribution shows relatively high frequencies for the middle colour grades, while the clarity distribution is concentrated mainly in the SI and VS categories.

4.3 Histogram Analysis

Histograms with density curves were plotted for the numerical variables.

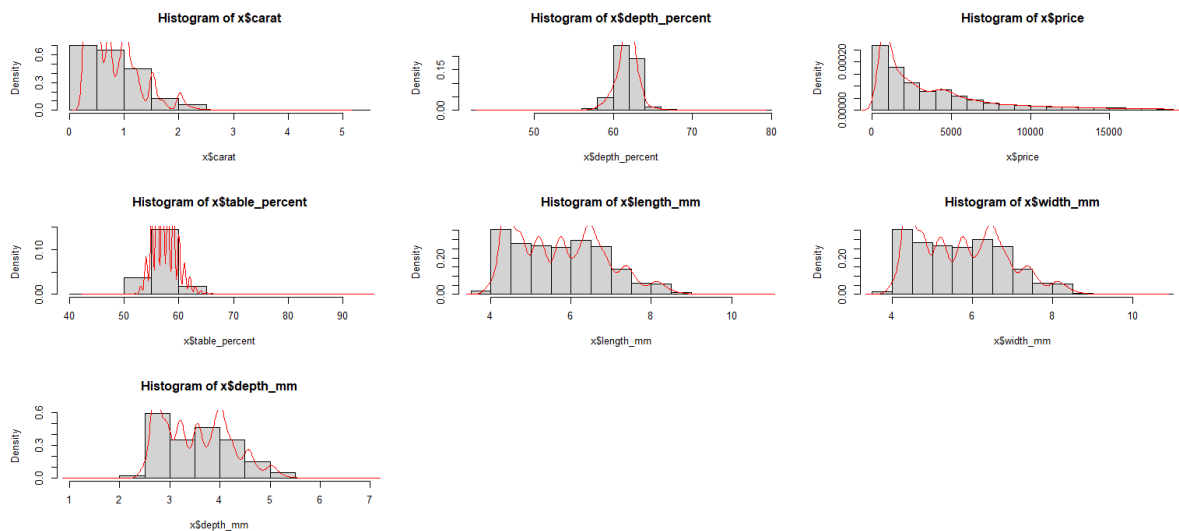


Figure 3: Histograms with density curves

The histogram of carat is positively skewed, with most observations concentrated at lower carat values. Price is strongly right-skewed, indicating that most diamonds have relatively lower prices while a smaller number of diamonds have substantially higher prices.

Depth percentage and table percentage are much more concentrated around their central ranges. Length, width, and depth show right-skewed patterns with some high-valued observations.

5 Relationship Analysis

5.1 Scatterplot Matrix

A scatterplot matrix was used to examine pairwise relationships among the numerical variables.

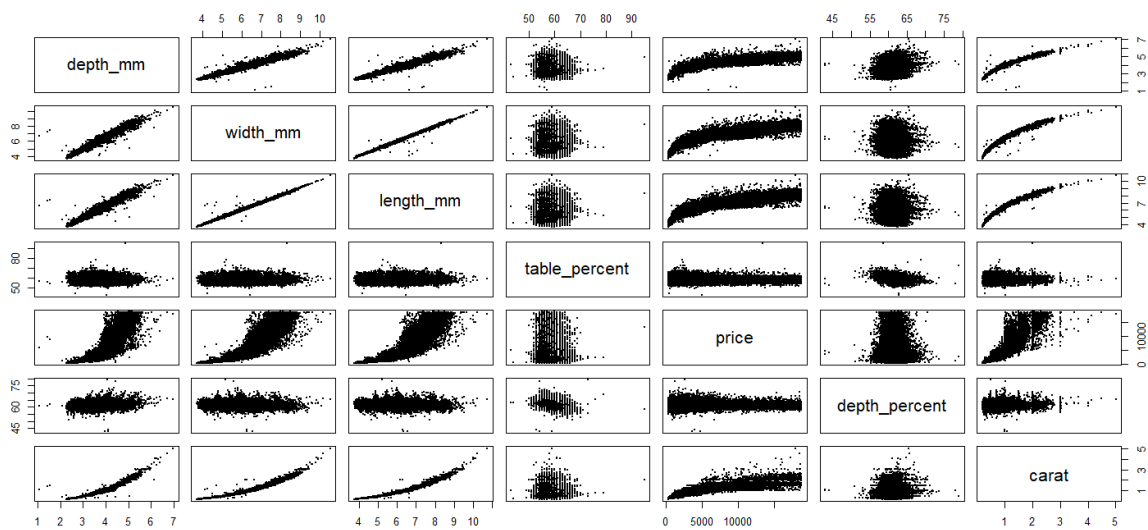


Figure 4: Scatterplot matrix of numerical variables

The scatterplot matrix shows very strong positive relationships among carat, length, width, and depth. The relationships between these physical measurements are almost linear in several cases.

The matrix also indicates a strong positive relationship between price and carat and strong positive relationships between price and the physical dimensions.

5.2 Price versus Carat

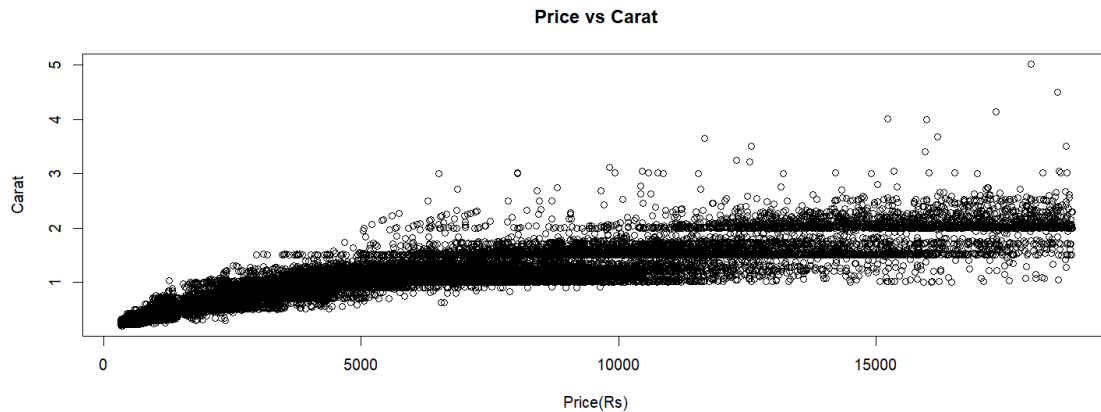


Figure 5: Relationship between price and carat

The scatterplot shows a strong positive association between price and carat. As carat increases, the price of a diamond generally increases. The relationship is not perfectly linear, and the variability also changes across the range of carat values.

The correlation matrix gives:

$$r_{\text{price,carat}} \approx 0.92.$$

Thus, carat has a very strong positive linear association with price.

5.3 Price versus Width

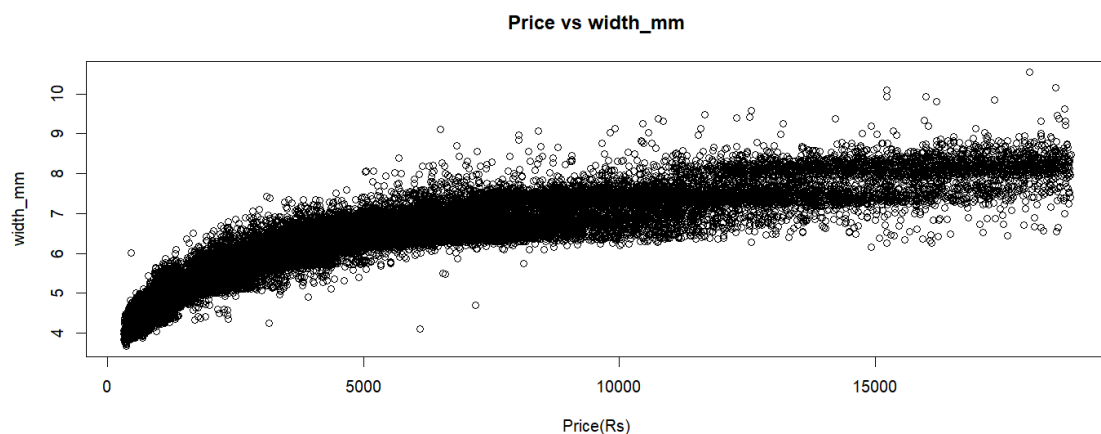


Figure 6: Relationship between price and width

The plot indicates a strong positive relationship between price and width. Larger diamonds in terms of width tend to have higher prices. The relationship is nonlinear to some extent, but the overall trend is clearly increasing.

The correlation matrix gives:

$$r_{\text{price,width}} \approx 0.89.$$

5.4 Price versus Depth

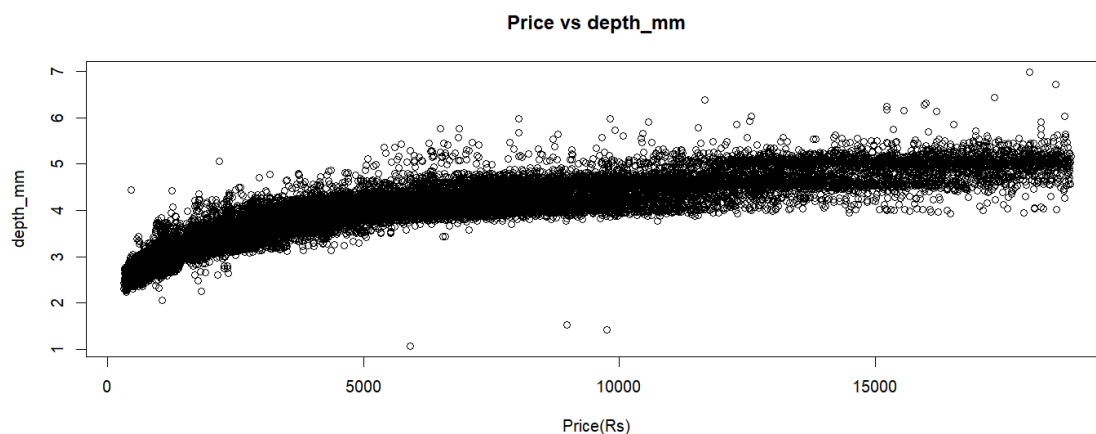


Figure 7: Relationship between price and depth

Price and depth also show a positive association. The relationship is weaker than the relationship between price and carat, but the overall scatterplot shows an increasing trend.

The correlation matrix gives:

$$r_{\text{price,depth}} \approx 0.88.$$

5.5 Price versus Length

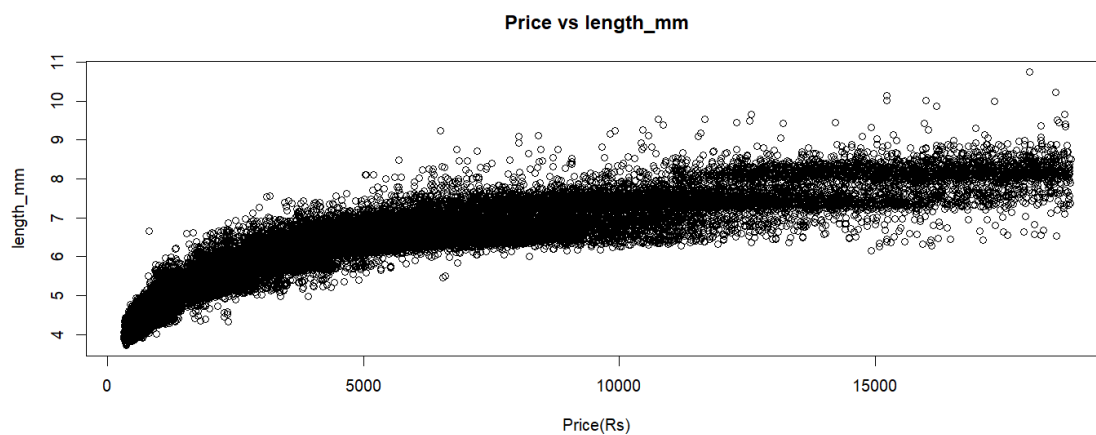


Figure 8: Relationship between price and length

A strong positive relationship is observed between price and length. Larger physical dimensions are generally associated with higher diamond prices.

The correlation matrix gives:

$$r_{\text{price,length}} \approx 0.89.$$

6 Correlation Analysis

The correlation matrix was calculated for the numerical variables in the dataset.

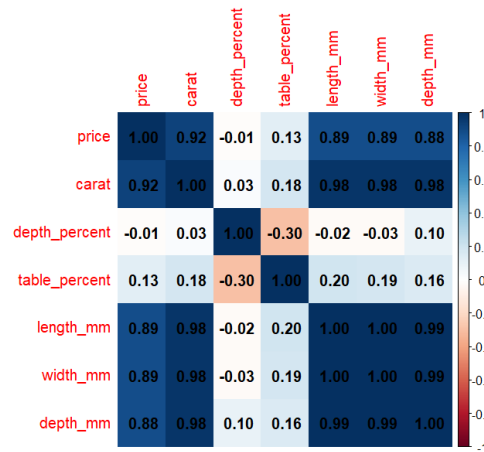


Figure 9: Correlation matrix of numerical variables

The supplied correlation plot gives the following approximate correlations:

Table 4: Important Correlation Coefficients

Variable Pair	Correlation	Interpretation
Price–Carat	0.92	Very strong positive
Price–Length	0.89	Strong positive
Price–Width	0.89	Strong positive
Price–Depth	0.88	Strong positive
Price–Table	0.13	Weak positive
Price–Depth percentage	-0.01	Very weak
Carat–Length	0.98	Very strong positive
Carat–Width	0.98	Very strong positive
Carat–Depth	0.98	Very strong positive
Length–Width	1.00	Extremely strong positive
Length–Depth	0.99	Extremely strong positive
Width–Depth	0.99	Extremely strong positive

The correlation analysis shows that carat and the physical dimensions of a diamond are highly correlated with price.

A particularly important observation is the very high correlation among length, width, depth, and carat. Such high correlations among predictors indicate the possibility of **multicollinearity** in a multiple regression model.

Depth percentage has almost no linear correlation with price in the supplied correlation matrix, while table percentage has only a weak positive correlation with price.

7 Multiple Linear Regression

7.1 Model

A multiple linear regression model was fitted with diamond price as the response variable and the remaining variables as explanatory variables: cut, color, clarity.

The model was fitted using:

```
1 model <- lm(price ~ ., data = x)
2 summary(model)
```

The general form of the fitted model can be represented as:

$$\begin{aligned}\widehat{price} = & \beta_0 + \beta_1(carat) + \beta_2(cut) + \beta_3(color) + \beta_4(clarity) \\ & + \beta_5(depth_percent) + \beta_6(table_percent) + \beta_7(length_mm) \\ & + \beta_8(width_mm) + \beta_9(depth_mm) + \epsilon.\end{aligned}$$

Since cut, colour, and clarity are categorical variables, R represents their categories using indicator variables in the fitted regression model.

Note: The variables `cut`, `color`, and `clarity` have a natural ordering in the diamond grading system. However, in the present regression model, they are treated as unordered categorical factors by `lm()`. An ordered-factor representation using `ordered()` could be considered in future work to explicitly incorporate their ordinal structure.

7.2 Estimated Regression Coefficients

Table 5: Estimated Regression Coefficients

Variable	Estimate	p-value
Intercept	-5012.995	1.98×10^{-14}
Carat	11547.957	$< 2 \times 10^{-16}$
Cut (Good)	481.436	$< 2 \times 10^{-16}$
Cut (Ideal)	750.349	$< 2 \times 10^{-16}$
Cut (Premium)	718.155	$< 2 \times 10^{-16}$
Cut (Very Good)	617.756	$< 2 \times 10^{-16}$
Color (E)	-209.010	$< 2 \times 10^{-16}$
Color (F)	-267.983	$< 2 \times 10^{-16}$
Color (G)	-476.426	$< 2 \times 10^{-16}$
Color (H)	-976.834	$< 2 \times 10^{-16}$
Color (I)	-1474.874	$< 2 \times 10^{-16}$
Color (J)	-2379.036	$< 2 \times 10^{-16}$
Clarity (IF)	5291.099	$< 2 \times 10^{-16}$
Clarity (SI1)	3636.587	$< 2 \times 10^{-16}$
Clarity (SI2)	2681.800	$< 2 \times 10^{-16}$
Clarity (VS1)	4543.200	$< 2 \times 10^{-16}$
Clarity (VS2)	4236.969	$< 2 \times 10^{-16}$
Clarity (VVS1)	4961.097	$< 2 \times 10^{-16}$
Clarity (VVS2)	4906.617	$< 2 \times 10^{-16}$
Depth (%)	60.464	2.46×10^{-10}
Table (%)	-24.766	$< 2 \times 10^{-16}$
Length (mm)	-1510.527	$< 2 \times 10^{-16}$
Width (mm)	1702.262	$< 2 \times 10^{-16}$
Depth (mm)	-2175.634	$< 2 \times 10^{-16}$

The estimated regression coefficients and their corresponding p-values are obtained directly from the regression summary. The estimated coefficients describe the direction and magnitude of the association between each explanatory variable and diamond price, while controlling for the other variables in the model.

The results show that all explanatory variables included in the model are statistically significant at the 5% level. Carat has a large positive estimated coefficient, indicating a strong positive association with diamond price. The categorical variables also show significant differences relative to their respective reference categories.

7.3 ANOVA Analysis

Table 6: Analysis of Variance (ANOVA) Table

Variable	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Carat	1	7.2799×10^{11}	7.2799×10^{11}	5.7631×10^5	$< 2 \times 10^{-16}$ ***
Cut	4	6.1279×10^9	1.5320×10^9	1.2128×10^3	$< 2 \times 10^{-16}$ ***
Color	6	1.2602×10^9	2.1003×10^8	1.6627×10^2	$< 2 \times 10^{-16}$ ***
Clarity	7	3.8352×10^{10}	5.4789×10^9	4.3373×10^3	$< 2 \times 10^{-16}$ ***
Depth (%)	1	5.2695×10^6	5.2695×10^6	4.1716	0.04111
Table (%)	1	9.2127×10^7	9.2127×10^7	7.2927×10^1	$< 2 \times 10^{-16}$ ***
Length (mm)	1	3.4804×10^9	3.4804×10^9	2.7552×10^3	$< 2 \times 10^{-16}$ ***
Width (mm)	1	1.4015×10^9	1.4015×10^9	1.1095×10^3	$< 2 \times 10^{-16}$ ***
Depth (mm)	1	2.7823×10^8	2.7823×10^8	2.2026×10^2	$< 2 \times 10^{-16}$ ***
Residuals	53893	6.8077×10^{10}	1.2632×10^6		

Significance codes: 0 “****” 0.001 “***” 0.01 “**” 0.05 “.” 0.1 “ ” 1. The ANOVA results show that most of the explanatory variables have a statistically significant effect on diamond price. Carat, cut, color, clarity, table percentage, length, width, and depth in millimetres are highly significant, with $p < 2 \times 10^{-16}$.

The variable depth percentage is also statistically significant at the 5% level, with a p-value of 0.04111.

Among the explanatory variables, carat has a particularly large F-statistic, indicating that it explains a substantial amount of the variation in diamond price. Overall, the ANOVA results demonstrate that the selected characteristics contribute significantly to explaining variations in diamond prices.

7.4 Model Fit

The fitted model has:

$$R^2 = 0.9206$$

and

$$\text{Adjusted } R^2 = 0.9205.$$

Therefore, approximately 91.98% of the variation in diamond prices is explained by the variables included in the model.

The overall F-statistic is:

$$F = 27159$$

with 23 and 53893 degrees of freedom.

The corresponding p-value is less than 2.2×10^{-16} , indicating that the overall regression model is statistically significant.

These values are obtained directly from the regression summary.

The following R commands were used to obtain these quantities:

```
1 summary(model)
2 anova(model)
3 coef(summary(model))
```

This avoids reporting numerical regression results that are not visible in the supplied graphs.

8 Regression Diagnostics

Standard regression diagnostic plots were generated to assess the adequacy of the fitted multiple linear regression model.

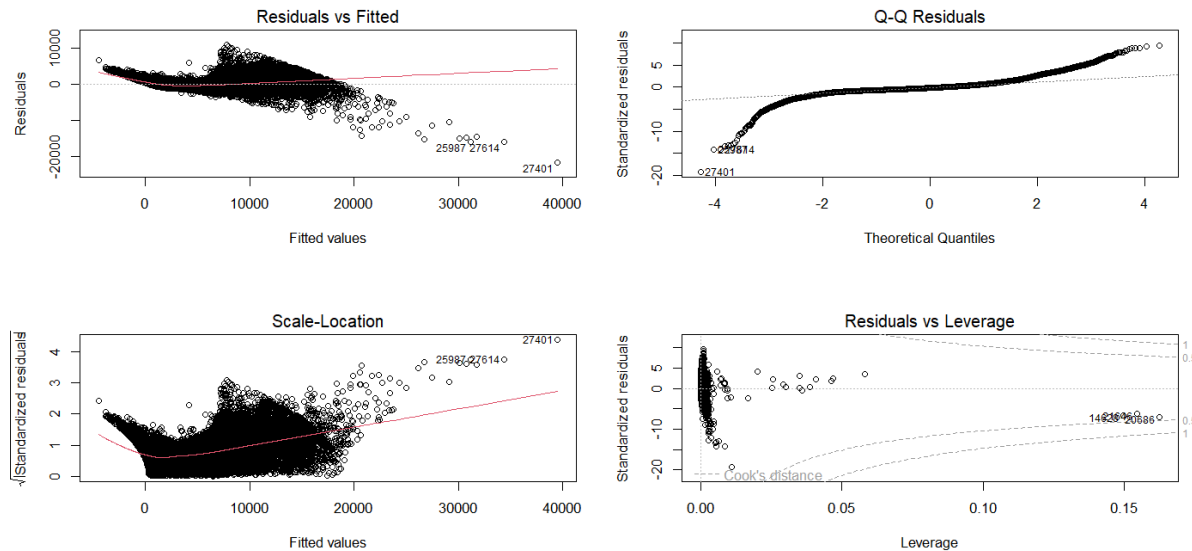


Figure 10: Regression diagnostic plots

The four standard diagnostic plots are:

1. Residuals vs Fitted
2. Normal Q-Q
3. Scale-Location
4. Residuals vs Leverage

8.1 Residuals vs Fitted

The Residuals vs Fitted plot is used to assess the linearity assumption. Ideally, the residuals should be randomly scattered around the zero line.

In the supplied diagnostic plot, the residuals show a noticeable pattern rather than being perfectly randomly distributed. The smooth red line also shows some systematic movement. This suggests that the linear model does not capture all nonlinear features of the relationship.

There are also several observations with unusually large positive or negative residuals.

8.2 Normal Q–Q Plot

The Normal Q–Q plot is used to examine whether the residuals are approximately normally distributed.

The supplied Q–Q plot shows clear departures from the reference line, particularly in the tails. Therefore, the residuals do not follow a perfectly normal distribution.

The extreme tail observations indicate the presence of unusual observations and heavy-tail behaviour.

8.3 Scale–Location Plot

The Scale–Location plot is used to assess whether the variance of the residuals is approximately constant.

In the supplied plot, the spread of the standardized residuals changes across the fitted values and the smooth red line shows an upward tendency. This provides evidence of non-constant variance, or heteroscedasticity.

8.4 Residuals vs Leverage

The Residuals vs Leverage plot is used to identify observations that may have a strong influence on the fitted regression model.

Most observations are concentrated at relatively low leverage, but a few observations have noticeably higher leverage. Some of these observations are also associated with large standardized residuals.

Therefore, potentially influential observations should be examined carefully when interpreting the fitted regression model.

8.5 Overall Assessment

Overall, the diagnostic plots indicate that the multiple linear regression model captures the major relationship between the predictors and diamond price, but the classical regression assumptions are not perfectly satisfied.

The main issues visible in the diagnostic plots are departures from normality, changing residual variance, and the presence of some unusual or potentially influential observations.

In addition, the correlation analysis shows very strong relationships among several explanatory variables. Therefore, multicollinearity is an important issue to consider in interpreting the individual regression coefficients.

9 Explanation of Results

The exploratory and regression analyses provide several important observations about diamond prices.

9.1 Effect of Carat

Carat has the strongest simple correlation with price among the main numerical variables considered, with a correlation of approximately 0.92.

The scatterplot clearly shows that larger-carat diamonds generally have higher prices. This indicates that carat is an important factor associated with diamond price.

9.2 Effect of Physical Dimensions

Length, width, and depth all show strong positive relationships with price. Their correlations with price are approximately 0.89, 0.89, and 0.88 respectively.

This indicates that larger physical dimensions are generally associated with higher diamond prices.

However, these variables are also extremely highly correlated with one another and with carat. Therefore, their separate regression coefficients may be affected by multicollinearity.

9.3 Effect of Depth Percentage

Depth percentage has an approximately zero simple correlation with price:

$$r \approx -0.01.$$

Therefore, there is almost no linear association between depth percentage and price when considered by itself.

9.4 Effect of Table Percentage

Table percentage has a weak positive correlation with price:

$$r \approx 0.13.$$

Thus, table percentage has a much weaker simple linear relationship with price than carat and the physical dimensions.

9.5 Effect of Cut, Colour and Clarity

Cut, colour, and clarity are categorical quality characteristics. Their inclusion in the regression model allows differences in diamond quality to be represented statistically.

The barplots show that the categories are not equally represented in the dataset. Hence, the fitted model uses reference categories and indicator variables to estimate the differences among the levels.

10 Conclusion

This study examined a diamond dataset using exploratory data analysis, correlation analysis, multiple linear regression, and regression diagnostics.

The exploratory analysis showed that price and carat are positively skewed and that several numerical variables contain outlying observations. The categorical variables cut, colour, and clarity also have different frequency distributions.

The scatterplots and correlation analysis revealed a very strong positive relationship between diamond price and carat. Price also has strong positive relationships with length, width, and depth. The physical dimensions themselves are highly correlated with carat and with one another.

A multiple linear regression model was then fitted with price as the response variable and the remaining variables as explanatory variables. The model provides a statistical framework for studying how diamond characteristics are associated with price.

However, the diagnostic plots show departures from normality, non-constant residual variance, and some unusual or potentially influential observations. The very high correlations among several predictors also indicate the presence of multicollinearity.

Overall, the analysis demonstrates how statistical and machine learning techniques can be used to understand and predict diamond prices from their physical and quality characteristics.

11 Future Scope

The analysis can be extended in several ways:

- Apply transformations such as a logarithmic transformation of price to reduce skewness.
- Investigate multicollinearity using Variance Inflation Factor (VIF).
- Compare multiple linear regression with decision trees and random forests.
- Divide the dataset into training and testing sets for proper prediction assessment.

- Use cross-validation to obtain a more reliable estimate of predictive performance.
- Investigate whether removing or treating extreme observations improves model performance.

12 R Code Used for Analysis

The following R code represents the main analysis performed in this project.

```
1 # Import data
2 x <- read.csv(file.choose(), header = TRUE)
3
4 # View data
5 head(x)
6 summary(x)
7 names(x)
8
9 # -----
10 # Boxplots
11 # -----
12
13 par(mfrow=c(3,3))
14
15 boxplot(x$carat,
16         main="carat",
17         ylab="Carat (ct)")
18
19 boxplot(x$depth_percent,
20         main="depth_percent",
21         ylab="(%) ")
22
23 boxplot(x$table_percent,
24         main="table_percent",
25         ylab="(%) ")
26
27 boxplot(x$price,
28         main="price",
29         ylab="(Rs.) ")
30
31 boxplot(x$length_mm,
32         main="length_mm",
33         ylab="(mm) ")
```

```
34
35 boxplot(x$width_mm,
36         main="width_mm",
37         ylab="(mm) ")
38
39 boxplot(x$depth_mm,
40         main="depth_mm",
41         ylab="(mm) ")
42
43 # -----
44 # Frequency tables
45 # -----
46
47 table(x$cut)
48 table(x$color)
49 table(x$clarity)
50
51 # -----
52 # Barplots
53 # -----
54
55 par(mfrow=c(1,3))
56
57 barplot(table(x$cut),
58         main="cut",
59         ylab="condition")
60
61 barplot(table(x$color),
62         main="color",
63         ylab="grade")
64
65 barplot(table(x$clarity),
66         main="clarity",
67         ylab="Condition")
68
69 # -----
70 # Histograms with density curves
71 # -----
72
73 par(mfrow=c(3,3))
74
```

```
75 hist(x$carat,
76     main="Histogram of x$carat",
77     xlab="x$carat",
78     probability=TRUE)
79
80 lines(density(x$carat), col="red")
81
82 hist(x$depth_percent,
83     main="Histogram of x$depth_percent",
84     xlab="x$depth_percent",
85     probability=TRUE)
86
87 lines(density(x$depth_percent), col="red")
88
89 hist(x$price,
90     main="Histogram of x$price",
91     xlab="x$price",
92     probability=TRUE)
93
94 lines(density(x$price), col="red")
95
96 hist(x$table_percent,
97     main="Histogram of x$table_percent",
98     xlab="x$table_percent",
99     probability=TRUE)
100
101 lines(density(x$table_percent), col="red")
102
103 hist(x$length_mm,
104     main="Histogram of x$length_mm",
105     xlab="x$length_mm",
106     probability=TRUE)
107
108 lines(density(x$length_mm), col="red")
109
110 hist(x$width_mm,
111     main="Histogram of x$width_mm",
112     xlab="x$width_mm",
113     probability=TRUE)
114
115 lines(density(x$width_mm), col="red")
```

```
116
117 hist(x$depth_mm,
118       main="Histogram of x$depth_mm",
119       xlab="x$depth_mm",
120       probability=TRUE)
121
122 lines(density(x$depth_mm), col="red")
123
124 # -----
125 # Scatterplot matrix
126 # -----
127
128 u <- subset(x,
129             select=c(price,carat,depth_percent,
130                      table_percent,length_mm,
131                      width_mm,depth_mm))
132
133 pairs(u)
134
135 # -----
136 # Individual scatterplots
137 # -----
138
139 plot(x$price, x$carat,
140      main="Price vs Carat",
141      xlab="Price(Rs)",
142      ylab="Carat")
143
144 plot(x$price, x$width_mm,
145      main="Price vs width_mm",
146      xlab="Price(Rs)",
147      ylab="width_mm")
148
149 plot(x$price, x$depth_mm,
150      main="Price vs depth_mm",
151      xlab="Price(Rs)",
152      ylab="depth_mm")
153
154 plot(x$price, x$length_mm,
155      main="Price vs length_mm",
156      xlab="Price(Rs)",
```

```
157     ylab="length_mm")
158
159 # -----
160 # Correlation matrix
161 # -----
162
163 cor(u, use="complete.obs")
164
165 # Optional corrplot
166 # install.packages("corrplot")
167 library(corrplot)
168
169 corrplot(cor(u, use="complete.obs"),
170          method="color",
171          addCoef.col="black")
172
173 # -----
174 # Multiple Linear Regression
175 # -----
176
177 model <- lm(price ~ ., data=x)
178
179 summary(model)
180
181 # ANOVA
182 anova(model)
183
184 # Regression coefficients
185 coef(summary(model))
186
187 # -----
188 # Diagnostic plots
189 # -----
190
191 par(mfrow=c(2,2))
192 plot(model)
193
194 # -----
195 # Additional model information
196 # -----
197
```

```
198 confint(model)
199
200 fitted(model)
201 residuals(model)
202
203 -----
```

GitHub Repository:

<https://github.com/supriyosaha7/Diamond-Price-Analysis>